

AdaBoost Step by Step

X_1	X_2	Y	Weight	Y -Pred	$W_{t,i}$	Normalized wt
3	7	1	0.2	1	0.16	0.166
2	9	0	0.2	1	0.24	0.25
1	4	1	0.2	0	0.24	0.25
9	8	0	0.2	0	0.16	0.166
3	7	0	0.2	0	0.16	0.166
					<u>0.96</u>	<u>1</u>

$n = 5$, Assign initial weight $= \frac{1}{n} = \frac{1}{5} = 0.2$

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - \text{error}}{\text{error}} \right)$$

$$\text{error} = 0.2 + 0.2 = 0.4$$

$$\therefore \alpha_1 = \frac{1}{2} \ln \left(\frac{1 - 0.4}{0.4} \right) = 0.2$$

For misclassified

$$\text{new_wt} = \text{current_wt} \times e^{\alpha_1}$$

For correctly classified

$$\text{new_wt} = \text{Current_wt} \times e^{-\alpha_1}$$

$$\therefore \text{wt for misclassified} = 0.2 \times e^{0.2} = 0.24$$

$$\text{wt for correctly classified} = 0.2 \times e^{-0.2} = 0.16$$

This means wt has been increased for misclassified points means on those points attention needs to be paid.

X_1	X_2	Y	New wt	Range	5 random no. bet 0-1
3	7	1	0.166	0 - 0.166	← 0.13
2	9	0	0.25	0.166 - 0.416	0.43
1	4	1	0.25	0.416 - 0.666	0.62
9	8	0	0.166	0.666 - 0.832	0.50
	7	0	0.166	0.832 - 1.0	0.8

Selecting rows 1, 3, 3, 3, 4 as max numbers fall between those ranges & creating new DF.

And this is called as upsampling

Then again α , new_wts are calculated

This process is repeated till number of stumping & then finally probability is calculated

$$p = (\alpha_1 \times h_1(x) + \alpha_2 \times h_2(x) \dots \alpha_n h_n(x))$$